

Study with Confucius: An AI-Based Immersive Educational Game with Multiple Educational Modes

Hanzhong Luo*

Interactive Media Design and
Technology Center, Tsinghua
University
China
luohz23@mails.tsinghua.edu.cn

Fengsen Gao*

Interactive Media Design and
Technology Center, Tsinghua
University
China
gfs22@mails.tsinghua.edu.cn

Ke Fang

Interactive Media Design and
Technology Center, Tsinghua
University
China
fang.ke@sz.tsinghua.edu.cn

Dejian Liu

Shenzhen International Graduate
School, Tsinghua University
China
liudj@nd.com.cn

Ziyun Lin

School of Design, South China
University of Technology
China
carameloko616@gmail.com

Wai Kin (Victor) Chan[†]

Tsinghua-Berkeley Shenzhen
Institute, Shenzhen International
Graduate School, Tsinghua University
China
chanw@sz.tsinghua.edu.cn

Abstract

Current history education faces challenges in cultivating student immersion and knowledge retention. This study introduces an innovative educational game, "Study with Confucius", which blends traditional Chinese cultural education with AI technology. The game's objective is to enhance student engagement and learning outcomes by simulating historical figures and reconstructing historical backgrounds. Utilizing ChatGPT technology, the game assigns distinct educational abilities to different Agents, resulting in three unique educational modes. Players interact with these Agents, learning classical knowledge and gaining insights into the associated wisdom and moral culture. User study findings demonstrate that the game improves student immersion and their understanding and memorization of knowledge. This game offers adolescents a novel learning approach, where they can engage with and inherit traditional culture through interactive and enjoyable means.

CCS Concepts

• Human-centered computing; • Interaction design; • Empirical studies in interaction design;

Keywords

History education, immersion, educational game, agent

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*Equal contribution.

[†]Corresponding author.

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1 Introduction

The main challenges faced by contemporary history education include student engagement and knowledge retention. Research by [Milam, 2021] shows that students generally find traditional history classes to be boring, a phenomenon observed worldwide. While augmented reality tools can enhance student engagement and long-term memory [Challenor and Ma, 2019], they are costly and difficult to implement. Additionally, [Yue and Zin, 2009] notes the limitations of these methods in fostering lasting and profound historical understanding. Immersion is crucial in history education. Research by [Buchanan and Palmer, 2017] shows a strong correlation between student immersion and the effectiveness of history learning. When students feel immersed, their understanding and interest in history significantly increase.

Educational games are regarded as an innovative and effective method for history education. Over the past decade, with the rapid development of video games, games designed primarily for educational purposes have emerged. As a significant component of serious games [Laamarti et al., 2014], educational games have gained considerable attention from both academic and industry sectors. Numerous studies have confirmed the effectiveness of educational games in various educational settings, ranging from early childhood to higher education [Backlund and Hendrix, 2013], and across multiple disciplines, including computer science and medical education [Li and Li, 2022; Zhonggen, 2019]. [Mozelius, 2014] noted that gamified environments significantly enhance student curiosity and engagement. Research by [Bado, 2022] indicates that gamified design not only increases engagement but also translates into tangible learning outcomes, improving overall learning effectiveness. [Plass et al., 2015] explored how gamified learning stimulates intrinsic motivation and self-directed learning through reward mechanisms, challenge designs, and instant feedback. Challenging games help students learn [Hamari et al., 2016]. Furthermore, [Rondon et al.,

2013] confirmed that gamified learning not only makes learning more enjoyable but also enhances knowledge acquisition and long-term memory retention.

In modern game design, as an advanced artificial intelligence (AI) system that surpasses basic text generation capabilities, the role of AI agents has become increasingly significant. According to [Petrović, 2018], AI agents enhance game interactivity and realism by equipping characters with autonomous decision-making and response capabilities through intelligent algorithms. Concurrently, the functionality and usability of large language models, such as ChatGPT, have been expanding, opening new possibilities for AI in gaming. The context-aware dialogue generation capabilities of language models can increase player engagement with NPCs [Csepreg, 2021], and open-ended conversations facilitated by these models may enhance player experience [Cox and Ooi, 2023; Gao et al., 2023]. An exemplary application of AI agents is the Stanford Town project described by [Park et al., 2023], which uses AI agents to simulate complex social interactions and character behaviors, creating highly interactive virtual environments. This not only enhances player immersion but also offers new avenues for game-based learning and simulation, demonstrating AI agents' potential to increase the value and interactivity of educational games.

In this study, AI agents are used to simulate historical figures and reconstruct historical contexts, creating a more interactive and immersive history learning environment. We anticipate that this approach will enhance students' immersion and deepen their understanding and retention of historical content. Based on this concept, we developed an educational game called "Study with Confucius", designed for teenagers. The game aims to teach ancient Chinese literature and traditional culture through gamified learning. Players take on the role of a disciple of Confucius during the Spring and Autumn period, interacting with various AI agents to accomplish tasks and gradually acquire knowledge of ancient Chinese texts and moral teachings. Leveraging ChatGPT's technological advantages, the game incorporates educational and gaming capabilities, enabling NPCs to function as AI agents. By combining these agents, three unique educational modes were created, each corresponding to a level where players learn and enhance their understanding while completing tasks.

This paper presents a detailed account of the design and implementation of the "Study with Confucius" game and evaluates its potential in the educational domain through user study analysis. We aspire to offer teenagers a novel learning method through this game, providing a delightful gaming experience that helps to transmit and celebrate traditional Chinese culture.

2 Game Design and Development

"Study with Confucius" (Figure 1) is an educational game designed for teenagers. In the game, players take on the role of a disciple of Confucius during the Spring and Autumn period, controlling their character's movements by clicking on the ground and interacting with NPCs through clicks. We employed ChatGPT to design and implement various educational and gaming features, assigning these features to different NPCs, thus transforming them into AI Agents. By combining these AI Agents, each with unique abilities (Figure 2), we developed three distinct educational modes. The

game is segmented into three levels, each tailored to a specific educational mode with unique gameplay, where players acquire knowledge by completing designated tasks.

The fundamental gameplay logic was implemented using the Unity game engine, while the capabilities of the AI Agents were realized through Prompt Graph. This approach allows for the breakdown of a capability into multiple smaller prompt function blocks, some of which are reusable. When creating new capabilities, we only need to combine existing prompt function blocks into a chain, greatly improving development efficiency.

2.1 "Two Children Argue About the Sun"

The first educational mode features one AI Agent that teaches the player, while another Agent assesses the player's comprehension. The initial level reconstructs the scene from the ancient Chinese text "The Debate on the Sun," where two children debate the distance of the sun, with Confucius unable to resolve their disagreement. In this scene, there are two children and Confucius; the child Agents handle teaching, and the Confucius Agent handles verification. The objective of this level is to immerse players in the interaction between the characters, helping them understand the key terms and the story's core message.

At the start of the level, players encounter the Confucius Agent, who discerns that they haven't yet engaged with "The Debate on the Sun" and directs them to find the two children. The child Agents teach by repeating their arguments from the text, requiring players to translate key terms into modern Chinese. The child Agents then use classical translation skills to compare the player's translations with the correct ones, allowing passage only if the translations are close enough. Subsequently, the player, as a disciple, needs to understand and convey "The Debate on the Sun" to the Confucius Agent, who uses intelligent judgment to compare the player's rendition with the correct one, correct mistakes, and answer questions, thus ensuring the player has mastered the knowledge and passed the level. Given that player input is free-form, restrictions are placed to maintain dialogue relevance with Confucius. If players stray from the topic, Confucius redirects them back. This mechanism is used for all free-dialogue Agents in the game.

2.2 "Jian Jia"

The second educational mode features two interrelated AI Agents, requiring the player to interact with both multiple times to learn the material. "Jian Jia" is an ancient Chinese love story, and the game's second level recreates this narrative. The player interacts with two main characters, the Dazhuang Agent and the Ajiao Agent, to gradually understand their story and learn about the core elements and ancient Chinese marriage customs.

Both main character Agents possess abilities such as dialogue, intelligent judgment, receiving items, crafting items, and driving the narrative. When players give items to them, the Agents use their intelligent judgment to determine if the items are meant for them or materials for crafting other items, thus deciding whether to receive the item or craft a new one. After the player completes tasks, the Agents collectively assess the player's progress and advance the storyline. The crafting ability also incorporates intelligent judgment; when players provide items and specify what they want



Figure 1: Game screen, a) game scene, b) "Two Children Argue About the Sun", c) "Jian Jia", d) "Zimo's Letters".

to craft, ChatGPT evaluates the feasibility of the request. Only logically sound crafting requests are executed. Besides the two character Agents, there are three environmental item Agents in the scene—reed, peach tree, and osprey—that only have the ability to generate items. Players obtain items by interacting with these Agents, which ChatGPT randomly generates based on their characteristics. Collecting these items is necessary to pass the level.

Upon entering the level, the player first encounters Dazhuang, who reveals his silent love for Ajiao across the river. However, his view is blocked by the reeds, preventing him from seeing her frequently. He asks the player to remove the reeds. After the player pulls up the reeds and gives them to Dazhuang, he crafts them into a small gift and asks the player to deliver it to Ajiao. Ajiao, in turn, gives the player a love knot to return to Dazhuang. Next, the player needs to shake the peach tree to obtain branches, which Dazhuang crafts into a mahogany hairpin. Finally, the player must throw objects at the osprey to obtain meat as a wedding gift for the couple. The level concludes with the marriage of Dazhuang and Ajiao, immersing the player in the story of "Jian Jia" and educating them about significant ancient Chinese wedding items like the love knot and mahogany hairpin.

2.3 "Zimo's Letters"

The third educational mode involves one agent posing specific questions, while the player discusses solutions with another agent, and the first agent evaluates the effectiveness of the proposed solutions. In the third level, we introduce a fictional disciple of Confucius named Zimo, who writes letters seeking help with various problems encountered during his travels. The player collaborates with the Confucius agent to discuss and devise solutions to the issues

presented in Zimo's letters. The Zimo agent possesses intelligent judgment and letter-writing capabilities, while the Confucius agent has an added recall ability, allowing him to reference previous topics and events to reinforce the player's memory of learned knowledge.

In this level, players not only enhance their classical Chinese reading skills but also internalize the famous saying from the Analects, "Diligent in learning and unashamed to ask questions." Upon entering this level, players first see a mailbox, which contains a letter from Zimo. In the letter, Zimo describes his struggle with drought management in a village, explaining that neither water storage nor well digging has resolved the issue, and thus he seeks advice from Confucius and the player. During discussions with the Confucius agent, direct answers are not provided; instead, the recall ability is used to reference the story of "Two Children Argue About the Sun," guiding players to propose the solution of humbly seeking advice from others. Players then draft a response to Zimo, who intelligently analyzes the content—if the response is unhelpful or off-topic, Zimo stops writing; if it aligns with the topic but is incorrect, Zimo explains why; and only a correct response about seeking advice is positively affirmed, indicating successful completion of the level.

"Study with Confucius" integrates traditional cultural education with interactive gaming through three distinct educational modes, aiming to enhance adolescents' interest in learning and knowledge retention. By combining AI technology with a game engine, we have achieved an efficient development process and created an engaging, educational virtual learning environment. These thoughtfully designed levels not only impart classical Chinese knowledge but also convey the wisdom and moral values of ancient times.

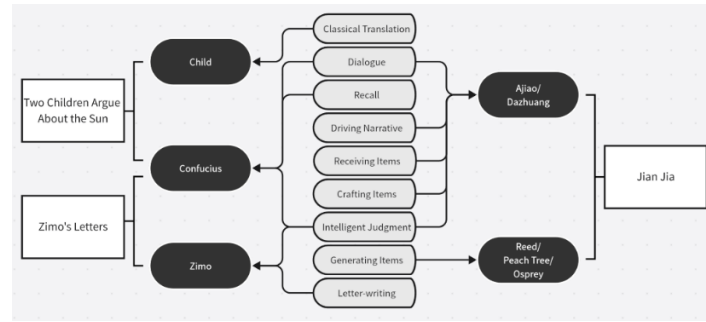


Figure 2: Agent and its ability contained in the three education modes respectively.

3 User Study

3.1 Study Design

The study consisted of both qualitative and quantitative experiments. The qualitative experiment recruited 32 eighth-grade students from various schools to uncover the strengths and weaknesses of the game, in order to improve the game and conduct a formal experiment. In the experiment, all students were randomly assigned to 9 small groups of 3-4 members, who simultaneously engaged in the game experience.

The quantitative experiment recruited 7 first-year high school students to evaluate the game experience and educational effectiveness. Students were randomly divided into two groups: a textbook learning group and a game learning group. The former group read learning materials, while the latter group played the game. Both groups immediately filled out a questionnaire and a knowledge test after completing their respective tasks.

3.2 Data Collection Methods

In the qualitative experiment (Figure 3a), researchers recorded the students' interactions and gameplay processes, and conducted semi-structured interviews after the experiment to collect post-gameplay interview data. The interview questions included, but were not limited to, the students' positive and negative experiences during the game, their memory of the learning content, and a comparison of the effectiveness with traditional textbook-based learning methods.

For the quantitative experiment (Figure 3b), the survey instrument consisted of the absorption dimension from the UWES-9S [Carmona-Halty et al., 2019] and the In-Game GEQ [IJsselsteijn et al., 2013]. The former measured absorption, which refers to the degree of the students' complete focus and pleasant immersion in the learning process, where higher levels indicate deeper engagement and greater difficulty in disengaging from the learning activity. The In-Game GEQ measured 7 dimensions within the game experience: Competence, Sensory and Imaginative Immersion, Flow, Tension, Challenge, Negative affect, and Positive affect. The knowledge test included tasks such as explaining classical Chinese vocabulary and recalling stories.



Figure 3: Experiment implementation, a) Qualitative experiment, b) Quantitative experiment.

4 Results

4.1 Qualitative Results

4.1.1 Immersion. Participants exhibited a high level of immersion, fully engaging in the experience. Despite initial distractions from mobile phones during the introduction, all participants actively joined in once the game commenced, taking turns playing and collectively strategizing to overcome various challenges. No one was distracted by unrelated activities. For instance, at the beginning of the game, Participant 1 from Group 6 was chatting with nearby classmates. However, as the game progressed, they became fully immersed. During interviews, most students found the game interface user-friendly, the controls simple, and the content engaging, which allowed them to focus entirely on learning.

4.1.2 Collaboration. Observation and interviews revealed a high level of collaboration among participants within each group, surpassing our initial expectations for the game's design, indicating the game's potential for promoting group discussion and cooperative learning. In the first part of the game, where participants interpreted ancient Chinese texts from "Two Children Argue About the Sun", they relied on memory and group discussions to complete the tasks. The second part of the game was the most engaging and time-consuming, as players enjoyed interacting with NPCs to create items. They crafted items like white wine and mahogany hairpins, leading to dynamic conversations. The third part of the game was the most challenging, requiring players to read letters written in classical Chinese by Confucius' disciples. Some players found these letters difficult to read but resolved their confusion through group discussions, eventually completing the tasks successfully.

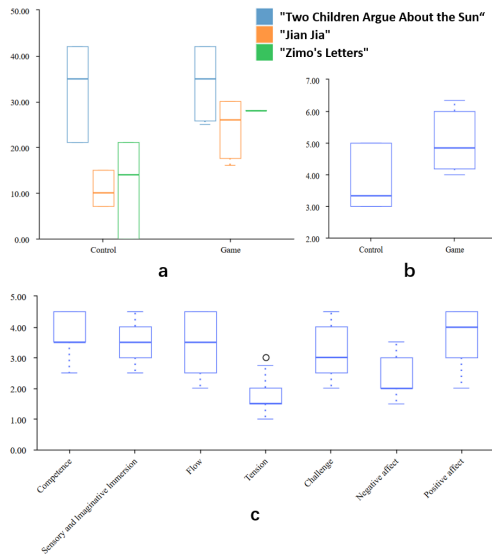


Figure 4: Quantitative Experiment Results, a) Knowledge Test Scores, b) Learning Absorption, c) Game Experience.

4.1.3 Learning Outcomes. We paid special attention to participants' understanding of literary works before and after playing the game. Students generally felt that the interactive tasks within the game enhanced their comprehension and retention of historical content. For example, regarding "Two Children Argue About the Sun", participants reported a deeper understanding of the ancient words and phrases that needed to be explained in the game. Participant 3 from Group 1 stated, "If this game could replace traditional classroom teaching, I would definitely choose this alternative. The learning outcomes would be better." Other participants echoed this sentiment, believing that the game provided a superior learning experience compared to the typical classroom environment. Participant 2 from Group 4 expressed anticipation for more levels, ideally covering all the classical texts in the curriculum, making learning easier for students.

4.2 Qualitative Results

4.2.1 Knowledge Assessment. The results of the knowledge assessment are shown in Figure 4a. During the "Jian Jia" and "Zimo's Letters" stages, students who learned through the game scored significantly higher than those who learned through textbooks. This indicates that agent interactions during these stages were particularly effective in enhancing learning outcomes. However, in the "Two Children Argue About the Sun" stage, there was no significant difference between the two groups. Based on observations of students' gameplay, we speculate that this may be due to the initial time students needed to adapt to the game's content and mechanics, which benefited their engagement and learning in subsequent stages.

4.2.2 Learning Absorption. The results for learning absorption are shown in Figure 4b. The median absorption score for the educational game group was significantly higher than that of the control

group (textbook learning group). This suggests that the game effectively increased students' immersion in the learning process. Qualitative feedback from players regarding their interactions with agents suggests that this method of learning made them more deeply engaged.

4.2.3 Game Experience. The results for game experience are shown in Figure 4c. Students reported high average scores for Competence, Sensory and Imaginative Immersion, and Positive Affect, indicating that they felt competent, deeply immersed, and positive while playing the game. Meanwhile, scores for Tension and Negative Affect were low, suggesting that the game effectively managed players' tension and negative emotions. The scores for Flow and Challenge were moderate, suggesting that while the game provided some level of flow experience, it also presented challenges due to high cognitive load, leading to varied participant experiences.

5 Conclusion and Future Work

This paper describes the design and implementation of the "Study with Confucius" game, which aims to provide classical Chinese and moral education. The game utilizes ChatGPT to design and implement multiple educational functions, which are combined through different AI agent configurations to create three distinct educational modes, resulting in three different gameplay levels. Players learn knowledge by completing specific tasks. To evaluate the game's experience and educational effectiveness, a user study was conducted, involving both qualitative and quantitative experiments. The qualitative experiment found that students displayed high levels of immersion and collaboration, with game interactions enhancing their understanding and memory of classical Chinese texts. The quantitative experiment showed that the educational game group significantly outperformed the traditional textbook learning group in knowledge test scores and learning absorption, with an overall positive game experience. However, the study had limitations such as a small sample size and high cognitive load. Future work will aim to simplify the game's operations, balance the difficulty, and improve the user experience to ensure that students from various backgrounds can easily engage with the game. Additionally, more classical Chinese levels will be added to cover a broader range of historical and cultural knowledge, making the game suitable for a wider age range. Finally, a larger sample size will be recruited for testing, and the "Study with Confucius" game will be implemented in schools to evaluate its effectiveness on students' learning outcomes and immersion.

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